UEC610: COMPUTER ARCHITECTURE

L	T	P	Cr
3	1	0	3.5

Course Objective: To introduce the state-of-the-art concept of parallelism followed in the modern RISC based computers by introducing the basic RISC based architecture. To make the students understand and implement various performance enhancement methods like memory optimization, Multiprocessor configurations, Pipelining and interfacing of I/O structures using interrupts and to enhance the student's ability to evaluate performance of these machines by using evaluation methods like CPU time Equation, To improve the student's skillset to program GPU clusters using CUDA language.

Syllabus

Fundamentals of Modern Computer Design: Introduction, Classes of Computers, Trends in Technology, Power, Energy and Cost in Integrated Circuits, Quantitative Principles of Computer Design, CISC and RISC architectures. Measuring, Reporting and Summarizing performance, Benchmarking, Principle of Locality, Amdahl's Law, Processor Performance Equation, Dependability (MTTF, MTTR).

Instruction Set Principles: Classes of Instruction set architecture- Stack, Accumulator,Register-Memory and Load-Store, Memory Addressing, Addressing modes, Operands, Types of operations in the instruction set, Addressing modes for Control-Flow, Data Transfer operations, Role of Compilers,

Pipelining and Parallelism: Idea of pipelining, Five stage Pipeline implementation of RISC instruction set, Performance Issues in Pipelining, Pipeline Hazards-Structural Hazards, Data hazards, Control Hazards, Performance of Pipelines with stalls, Performance evaluations with pipelining, Prediction Techniques, Static and Dynamic Branch Prediction, Branch Prediction Buffers, Pipeline Implementation of RISC-V Architecture, Exceptions in RISC-V, Pipeline Scheduling and Loop Unrolling.

Multi-Core processors: Flynn's multiprocessor architectures, Graphics Processing Units (GPUs), CPU vs GPU, GPU clusters for high performance, Introduction to CUDA programming, NVIDIA landscape for GPUs, CUDA programming examples Cache Coherence, Synchronization, Models of Memory Consistency-Snoopy Protocols, Brief introduction to DSP processors.

Memory Hierarchy Design: Introduction to Multilevel Memories, Principle of Temporal and Spatial Locality, Cache memory, Cache Addressing, Cache Organization, Write Policies, Reducing Cache Misses, Cache Associatively Techniques, Reducing Cache Miss Penalty, Reducing Hit Time, Main Memory Technology, Virtual memory and its addressing, Fast Address Translation, Page tables, Multilevel Page table, Translation Lookaside buffer, Crosscutting issues in the design of Memory Hierarchies.

Familiarization with standard: IEEE 754 for floating-point arithmetic, IEEE 802 for networking (Ethernet/Wi-Fi), IEEE 1516 for simulation (HLA), and IEEE 1471/42010 for architectural description.

Course Learning Objectives (CLO)

1. Apply the Amdahl's law and Processor Performance Equation to evaluate performance.
2. Characterize the processor instruction set architecture and addressing modes.
3. Identify the concept of parallelism and pipelining and hazard removal in context to the same.
4. To demonstrate GPU based system using CUDA programming
5. Investigate the Memory Hierarchy design, Memory Addressing and Address translation

Text Books

1. Hennessy, J. L., and D. A. Patterson. *Computer Architecture: A Quantitative Approach*, 2nd ed. San Mateo, CA: Morgan Kaufman, 1995. ISBN: 1558603727
2. Hennessy, J. L., Patterson, D. A., Computer Architecture: A Quantitative Approach, Elsevier (2009) 4th ed.

3. Hamacher, V., Carl, Vranesic, Z.G. and Zaky, S.G., Computer Organization, McGraw-Hill (2002) 2nd ed.
4. The CUDA Handbook: A Comprehensive Guide to GPU Programming, Nicholas Wilt (2013)

Reference Books

1. Murdocca, M. J. and Heuring, V.P., Principles of Computer Architecture, Prentice Hall (1999) 3rd ed.
2. Stephen, A.S., Halstead, R. H., Computation Structure, MIT Press (1999) 2nd ed.
3. Programming Massively Parallel Processors: A Hands-on Approach, 4th. edition, Wen-mei W. Hwu (2022)
4. Programming in Parallel with CUDA: A Practical Guide, Richard Ansorge (2022)
5. CUDA Programming: A Developer's Guide to Parallel Computing with GPUs, Shane Cook (2012)